EE4-57
EE9-CS2-2

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc and EEE/EIE PART IV: MEng and ACGI

Corrected copy

DISCRETE-EVENT SYSTEMS

Tuesday, 30 April 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : D. Angeli
Second Marker(s) : E.C. Kerrigan

© Imperial College London

1. Consider the Finite Deterministic Automaton (FDA) G whose transition diagram is shown in Fig. 1.1.

In particular, the set of events is $E = \{s, e, f, r\}$, denoting respectively start of operations, end of operations, fault events and successful recover of the machine. There are five states $X = \{0, 1, 2, 3, 4\}$, with 0 initial state and $X_m = \{0, 2\}$ the set of terminal states.

a) Let $\mathscr{L}(G|x)$ and $\mathscr{L}_m(G|x)$ denote, respectively, the language generated and marked by automaton G when x is the initial state. We say that $x_1 \sim x_2$ whenever $\mathscr{L}(G|x_1) = \mathscr{L}(G|x_2)$ and $\mathscr{L}_m(G|x_1) = \mathscr{L}_m(G_2|x_2)$. Find the set of equivalent states (equivalence classes) according to the equivalence relation $\sim$. [4]

b) Find the minimal finite deterministic automaton $G_{\min}$ that generates and marks the languages $\mathscr{L}(G)$ and $\mathscr{L}_m(G)$ respectively. [4]

c) Consider next two automata G_1 and G_2, representing faulty machines, with event sets $E_i = \{s_i, e_i, f_i, r_i\}$, $i = 1, 2$ respectively, and transition diagram shown in Fig. 1.2. Compute the parallel composition $G_1 || G_2$. [5]

d) Find an Finite Nondeterministic Automaton G_N that generates the language $P_F(\mathscr{L}(G_1||G_2))$, where $F = \{f_1, f_2, r_1, r_2\}$ and $P_F(\mathscr{L})$ denotes the projection of $\mathscr{L}$ over F. [3]

e) Find the deterministic observer automaton G_O that generates the same language as G_N. [4]

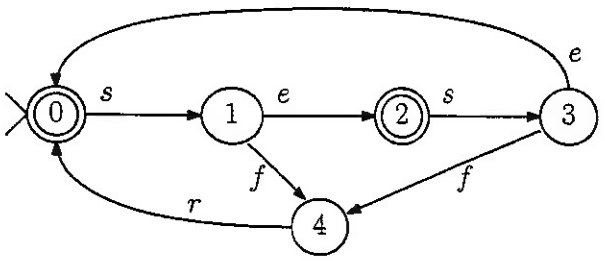


Figure 1.1 Automaton G: transition diagram.

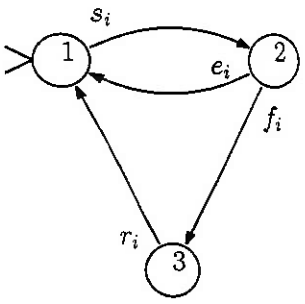


Figure 1.2 Automata G_i: transition diagram.

2. Two mirrors denoted as M1 and M2 are arranged as in Fig. 2.1. Each mirror can take 3 positions, respectively denoted V, D, H, as in Vertical, Diagonal and Horizontal. Moreover, mirror orientation can instantaneously be rotated by 45 degrees, when events c_i and a_i $(i = 1, 2)$ occur, corresponding respectively to clockwise and anticlockwise rotations of mirror i.

a) Model the system described above as a Finite Deterministic Automaton so as to keep track of the positions of the two mirrors and of the events occurring, assuming the initial position is Diagonal for both mirrors. [5]

b) A light source, which is always ON, is pointing towards mirror 1 as shown in the figure. Assuming that light is reflected by the two mirrors as in the figure, model the following 'informal' specification as a Finite Deterministic Automaton G_K: light should never bounce back on the Light Source. [5]

c) Assume next that uncontrollable events are $E_{uc} = \{c_2\}$. Verify if the specification above is controllable or uncontrollable with respect to E_{uc}. [5]

d) Determine the maximal set of uncontrollable events $E_{u\max}$ which keeps $\mathscr{L}(G_K)$ controllable. [5]

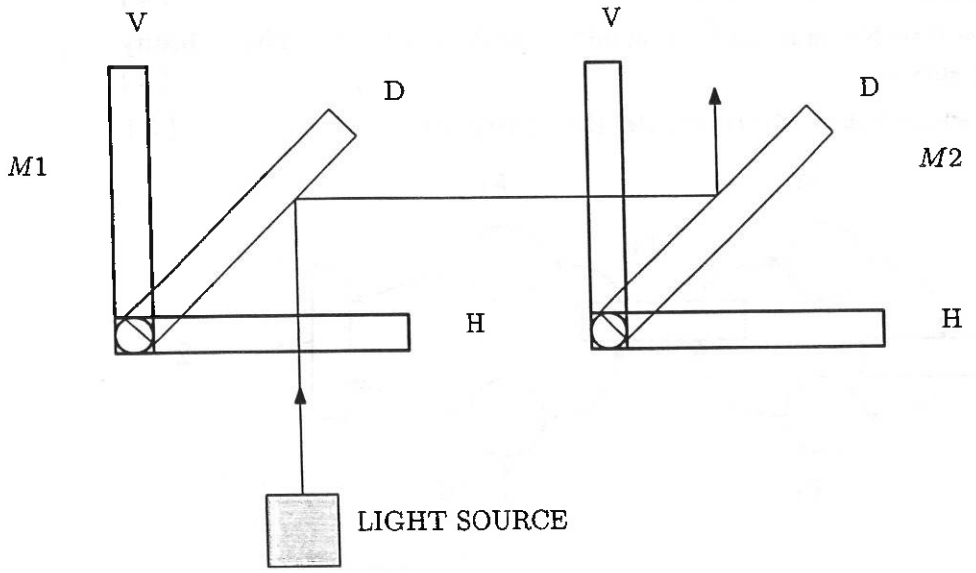


Figure 2.1 Mirror systems.

3. Consider the Petri Net N whose diagram is shown in Fig. 3.1. The associated net has 4 places $P = \{p_1, p_2, p_3, p_4\}$ and 3 transitions $T = \{t_a, t_d, t_x\}$. This net models the transit of people within an exhibition arranged over two rooms of finite capacity.

Transition t_a represents arrivals in room 1, t_d means departure from the first room and transit into the second room, while t_x stands for exit. It is assumed that one person, a guardian, will normally station in room 1 and welcome new visitors and that new people can only enter if the guardian is there.

a) Find the matrices *Pre*, *Post* and the incidence matrix C associated to the considered net. [3]

b) Describe the physical meaning in the net of the number of tokens in each of the four places p_1, p_2, p_3, p_4. [4]

c) Find the minimal P-semiflows of the net and discuss them in the light of the physical meaning of places as detailed in item b). [2]

d) Find the minimal T-semiflows of the net and discuss their meaning. [2]

e) Consider the initial marking $M_0 = [2, 1, 0, 2]$, find the reachable set $\mathscr{R}(M_0)$ and sketch the associated transition diagram. [4]

f) Is the Petri Net structurally bounded? Is the Petri Net conservative? (Justify your answer). [2]

g) Is the Petri Net (N, M_0) reversible? (Justify your answer). [3]

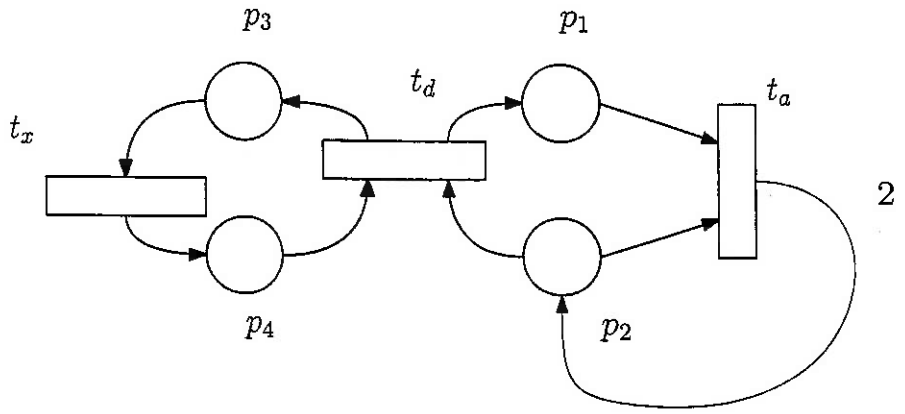


Figure 3.1 Petri net diagram

4. We model the availability of a fleet of Electric Vehicles (EVs) as a homogeneous Markov chain. Each vehicle can be in 3 possible states: C, charging at a docking station; I, fully charged and idle at a docking station; T, travelling. The transition rate between state C and I is assumed to be λ_1, while the transition rate between state T and C equals λ_3. Moreover, regardless of the current state of charge, the transition rate towards T is supposed to be λ_2.

a) Sketch the transition diagram of the associated Markov Chain. [3]

b) Is the Markov Chain ergodic? (Justify your answer) [2]

c) Write the Kolmogorov equations describing how the probability of being in each of the states evolve in time. [4]

d) Compute the steady-state probability distribution. [4]

e) Justify the following claim: "the average fraction of time spent travelling by each vehicle is independent of charging speed at the docking station". [3]

f) Compute the average time between two consecutive journeys. [4]